Term-minimal realizations of nine line-type configurations in quadratic mass-action systems

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Abstract

Consider two-species mass-action systems with reactant molecularity at most two, unrestricted products, and a positive equilibrium whose Jacobian is the zero matrix. We count nonzero monomials separately in the two component equations, in concentration coordinates. After moving the equilibrium to the origin for geometric analysis, we require the two homogeneous quadratic components to have nonzero resultant. For nine configurations of invariant-line types, the minimum term counts are $8, 8, 8, 8, 8, 8, 8, 6, 6$. Rational witnesses attain every bound. The lower bounds follow from a nonnegative six-parameter representation and an exhaustion of the eight support faces with fewer than eight terms. Relative to the examples in Yeung and Erban's classification paper, including its six-term example, five upper bounds decrease and four existing upper bounds are proved optimal within this domain. The count concerns ODE terms; minimum reaction counts are a separate problem.

1 The question and the result

Yeung and Erban [1] identify nine realizable geometric configurations at positive equilibria with zero linearization in planar second-order chemical systems. Their discussion asks whether the nonzero-term counts of the displayed examples can be reduced. This paper gives exact minima for the nine invariant-line configurations under the explicit nonzero-resultant condition below. Table 1 states the result and compares the examples.

The geometric background is the classification attributed to Lyagina [2] and developed in Newton's invariant-line description [3]. We use the line-type notation explained in Section 3, and distinguish configurations by their multisets of line types. This avoids depending on inconsistent numerical labels in the source examples. The proof below does not require a new proof of the historical classification or an identification of its entire domain with ours.

Configuration	Row[a]	Table 1	Best[b]	Minimum	Comparison
H	1	9	9	8	decreases
$\mathrm{P_2H}$	4	8	8	8	attained before
D, H	5	10	10	8	decreases
$\mathrm{D}, \mathrm{P_2H}$	8	8	8	8	attained before
$\mathrm{P_1}, \mathrm{P_2}, \mathrm{H}$	11	9	9	8	decreases
$\mathrm{P_1}, \mathrm{H}, \mathrm{H}$	12	9	9	8	decreases
$\mathrm{P_1}, \mathrm{P_2}, \mathrm{P_2H}$	14	8	8	8	attained before
$\mathrm{P_1}, \mathrm{H}, \mathrm{P_2H}$	15	8	6	6	attained before
$\mathrm{P_1}, \mathrm{P_2H}, \mathrm{P_2H}$	16	8	8	6	decreases

Table 1: Minimum numbers of component monomials. [a]Row label in Table 1 of [1], used only to locate the example. [b]Best count among that table and the six-term example in its Section 4. The configurations here are computed from the coefficients, not copied from the printed labels.

2 Chemical domain and the quantity minimized

Write $\dot{x} = f(x, y)$ and $\dot{y} = g(x, y)$, with real coefficients and total degree at most two. A reaction $\alpha X + \beta Y \longrightarrow \gamma X + \delta Y$ has a positive rate constant and $\alpha + \beta \leq 2$. We impose no bound on $\gamma + \delta$. Consequently the coefficient of a monomial not containing x must be nonnegative in f, and the coefficient of a monomial not containing y must be nonnegative in g. These sign conditions also suffice: realize each nonzero coefficient by a reaction that adds or removes one molecule of the corresponding species. A negative coefficient has that species in its reactant, so removal is permitted. This constructs a network with positive rate constants.

Let (x_*, y_*) be a selected positive equilibrium and require $Df(x_*, y_*) = Dg(x_*, y_*) = 0$. Positive changes of concentration units place it at $(1, 1)$. Such changes multiply each coefficient by a positive number and preserve the number of nonzero terms. With $u = x - 1$ and $v = y - 1$, the shifted equations have the form

$$\dot{u} = P(u, v) = Au^2 + Buv + Cv^2, \qquad \dot{v} = Q(u, v) = Du^2 + Euv + Fv^2. \tag{1}$$

Translation is used to study invariant lines; all term counts remain in the concentration variables x, y.

In the monomial order $(1, x, y, x^2, xy, y^2)$, the coefficient vectors are

$$\begin{aligned} a &= (A + B + C, -2A - B, -B - 2C, A, B, C), \\ b &= (D + E + F, -2D - E, -E - 2F, D, E, F). \end{aligned} \tag{2}$$

The chemical sign conditions give the following parameterization:

$$\begin{aligned} \theta &= (s, t, c, q, r, d) \in \mathbb{R}^6_{\geq 0}, \\ a &= (s, -2s - t, t, s + t + c, -t - 2c, c), \\ b &= (q, r, -2q - r, d, -r - 2d, q + r + d), \\ (A, B, C, D, E, F) &= (s + t + c, -t - 2c, c, d, -r - 2d, q + r + d). \end{aligned} \tag{3}$$

Every chemical quadratic system with the specified equilibrium and zero Jacobian has this form, since $s = a_{00}$, $t = a_{01}$, $c = a_{02}$, $q = b_{00}$, $r = b_{10}$ and $d = b_{20}$. Conversely, direct substitution in (3) verifies all these properties.

Definition 1. A parameter vector in (3) is admissible if the homogeneous resultant $\mathrm{Res}(P, Q)$ is nonzero. Its term count is

$$M(\theta) = \mathrm{nnz}(a) + \mathrm{nnz}(b).$$

For a specified line-type multiset T, its minimum is taken over all admissible real parameter vectors with type T.

The resultant condition means that P and Q have no common zero on the complex projective line; it excludes zero components and common factors. It is part of the domain, not a conclusion inferred from an isolated real equilibrium. The same monomial appearing in both components contributes two to M. Positive concentration and time scalings preserve M; arbitrary affine changes of concentration variables are not operations in this minimization problem. A reaction may contribute to both components, so M does not count reactions.

3 Invariant-line types

A finite slope $z = v/u$ defines an invariant line precisely when

$$h(z) = zP(1, z) - Q(1, z) = Cz^3 + (B - F)z^2 + (A - E)z - D = 0. \tag{4}$$

At any real root, $p(z) = P(1, z) \neq 0$ by the resultant condition. Set

$$n(z) = E + (2F - B)z - 2Cz^2, \qquad \rho(z) = \frac{n(z)}{p(z)} = 1 - \frac{h'(z)}{p(z)}. \tag{5}$$

For a simple root, the four symbols in Table 2 record the range of ρ. A double root has $\rho = 1$ and is denoted D. The multiplicity condition is essential: D does not include triple roots. All roots and the vertical direction are counted as projective directions, so each line appears once.

Symbol	P_1	P_2	H	P_2H
Condition (simple root)	$\rho > 1$	$0 < \rho < 1$	$\rho < 0$	$\rho = 0$

Table 2: Operational definitions of the four simple line types.

These quantities have a direct geometric meaning. Near a finite invariant line $v = zu$, put $w = v - zu$. Linearizing the scalar trajectory equation gives $dw/du = \rho w/u$, whose first variation is proportional to $|u|^\rho$. This distinguishes approach tangent to the line at the origin (P_1), relative approach at large distance (P_2), vanishing transverse distance at large distance (H), and the boundary case (P_2H). The definitions used in the proof are the algebraic conditions of (5), rather than sector counts alone.

The vertical line $u = 0$ is invariant if and only if $C = 0$. In this case $F > 0$ and $B \leq 0$, so it is a simple projective root and

$$\rho_\infty = B/F. \tag{6}$$

Indeed the swapped slope polynomial is $wQ(w, 1) - P(w, 1)$, whose derivative at $w = 0$ is $F - B > 0$. We use unordered multisets of these symbols. No assertion that all nine are distinct under arbitrary local topological conjugacy is needed.

4 Exact upper bounds

Theorem 2. *In the domain of Definition 1, the minima for the nine configurations in Table 1, in the displayed order, are*

$$8, 8, 8, 8, 8, 8, 8, 6, 6.$$

Table 3 gives integer parameter vectors that attain these counts. Together with (3), the table specifies every coefficient of every witness exactly. Positive common scaling of the vector amounts to a change of time units.

ID	Configuration	s	t	c	q	r	d	M
1	H	0	0	3	1	0	2	8
2	P_2H	1	0	0	2	0	1	8
3	D, H	0	0	9	4	0	2	8
4	D, P_2H	2	0	0	3	0	1	8
5	P_1, P_2, H	0	0	56	25	0	12	8
6	P_1, H, H	1	1	0	0	0	2	8
7	P_1, P_2, P_2H	31	0	0	33	0	64	8
8	P_1, H, P_2H	1	0	0	0	0	1	6
9	P_1, P_2H, P_2H	1	0	0	1	0	0	6

Table 3: Exact witnesses, with parameters in (3).

For completeness, Table 4 records their slope polynomials and transverse ratios. In row 5 the ratios follow the root order $2, 2/7, 3/8$. In rows 6–8 the paired signs have the same order in the roots and the ratios. The only real root in row 1 is 2, and row 2 has no real finite root. In rows 3 and 4, 1/3 and 1/2, respectively, are double roots.

ID	$h(z)$	Finite roots : ρ	ρ_∞
1	$(z-2)(3z^2-3z+1)$	$2 : -4/3$	—
2	$-3z^2+3z-1$	none	0
3	$(z-2)(3z-1)^2$	$2 : -16/9; \quad 1/3 : 1$	—
4	$-(2z-1)^2$	$1/2 : 1$	0
5	$(z-2)(7z-2)(8z-3)$	$2, 2/7, 3/8 : -25/14, 7/10, 48/35$	—
6	$-3z^2+6z-2$	$1 \pm 1/\sqrt{3} : 4 \pm 3\sqrt{3}$	$-1/2$
7	$-97z^2+159z-64$	$(159 \pm \sqrt{449})/194 : 1 \pm \sqrt{449}/31$	0
8	$-z^2+3z-1$	$(3 \pm \sqrt{5})/2 : 1 \pm \sqrt{5}$	0
9	$z-z^2$	$0 : 0; \quad 1 : 2$	0

Table 4: Algebraic checks of the line types. A dash means that the vertical direction is not invariant.

The homogeneous resultants, in the same order, are

$$9,\ 9,\ 1296,\ 64,\ 1960000,\ 4,\ 9042049,\ 1,\ 1.$$

All are nonzero. Thus each displayed system is admissible; substituting its parameters into (3) gives the asserted count and chemical realization. In row 6, direct substitution in (5) is also a convenient way to check the two finite types.

5 Lower bounds over real parameters

The lower bound does not depend on searching a finite grid of coefficients. It follows from the support of the nonnegative coordinates in (3). Encode the positive entries of (s, t, c) by a mask with weights $(1, 2, 4)$. Use the same weights for (q, r, d). Positivity and the absence of cancellation in the sums in (3) give the component counts

mask	1	2	3	4	5	6	7
component terms	3	4	5	3	5	5	6.

(7)

A zero mask would make one component identically zero and is excluded. It follows at once that $M \geq 6$.

Lemma 3. *An admissible system with $M < 8$ has configuration either $\{\mathrm{P_1}, \mathrm{H}, \mathrm{P_2H}\}$ or $\{\mathrm{P_1}, \mathrm{P_2H}, \mathrm{P_2H}\}$.*

Proof. Equation (7) leaves just eight pairs of masks:

$$(1,1), (1,4), (4,1), (4,4), (1,2), (2,1), (4,2), (2,4).$$

Three pairs have both $s = q = 0$, namely $(4,4), (4,2), (2,4)$; in these pairs, one has $P(1,1) = s = 0$ and $Q(1,1) = q = 0$. The components have a common projective zero, contradicting admissibility.

Interchanging the species exchanges the masks and preserves the line types. For a nonzero finite slope this follows by conjugating the trajectory slope map $f(z) = Q(1,z)/P(1,z)$ with $z \mapsto 1/z$; the derivative at its fixed point remains ρ. The horizontal and vertical charts give the

same statement for the remaining directions. Hence only $(1,1),(1,4),(1,2)$ need be considered. Write their positive coordinates as $a, b > 0$.

For $(1,1)$, $P = au^2$ and $Q = bv^2$. The directions are $0, a/b, \infty$ and the ratios are $0, 2, 0$. The configuration is $\{\mathrm{P}_1, \mathrm{P}_2\mathrm{H}, \mathrm{P}_2\mathrm{H}\}$.

For $(1,4)$, $P = au^2$ and $Q = b(u-v)^2$. Here

$$h(z) = -bz^2 + (a+2b)z - b, \qquad \Delta = a^2 + 4ab > a^2.$$

The finite roots have ratios $1 \pm \sqrt{\Delta}/a$ and the vertical direction has ratio zero. The configuration is $\{\mathrm{P}_1, \mathrm{H}, \mathrm{P}_2\mathrm{H}\}$.

For $(1,2)$, $P = au^2$ and $Q = bv(v-u)$. The directions $0, (a+b)/b, \infty$ have ratios $-b/a, 2 + b/a, 0$, again giving $\{\mathrm{P}_1, \mathrm{H}, \mathrm{P}_2\mathrm{H}\}$. In all three families the resultant is $a^2b^2 > 0$. This exhausts all eight faces for all real choices of their positive coordinates. □

Proof of Theorem 2. The last two configurations attain six terms by Table 3, and (7) proves that fewer are impossible. Lemma 3 excludes fewer than eight terms for each of the other seven configurations. Their eight-term witnesses complete the proof. □

6 Computation and assurance

The calculation that led to the witnesses decomposes the parameter cone into $7 \cdot 7 = 49$ nonempty support faces and considers the nine target types on each face. Polynomial solving supplies candidates. The minima depend only on the nine rational witnesses and the lower-bound argument in Section 5; unresolved feasibility cases on larger faces are not used in that argument.

For a rational witness, an exact certificate contains the component coefficients, a factorization of h, root multiplicities, rational isolating intervals, and signs of $p, n, n-p$ at each real root. A checker using rational arithmetic verifies the factorization, squarefreeness of its factors, Sturm root counts, interval signs or exact zero identities, the vertical direction, and the resultant. It also checks the equilibrium, zero Jacobian, chemical signs, term count, and a reaction realization. Irrational roots of rational polynomials require isolating intervals, not floating-point classification. Lower-bound certificates encode the component support counts and the three symbolic families in Lemma 3.

The assurance consists of these exact certificate checks and the displayed algebraic proof; no proof-assistant formalization or human peer review is claimed. Provenance, checker versions, and the bounded literature search are recorded separately in the accompanying `VERIFICATION-RECORD.md`.

7 Independent verification

The nine minima were also obtained in a separate verification that used the problem specification and its stated definitions, while withholding the research-side minima, proofs, implementations, and certificates. It constructed integer-coefficient witnesses and derived a lower bound by classifying all $2^6 = 64$ zero-or-positive supports of the nonnegative parameters. Removing supports with a zero component leaves the 49 faces considered above. The eight supports with $M < 8$ reduce to the same three positive parameter families after excluding common factors and interchanging the species. All nine independently obtained minima agree with Theorem 2.

The separate derivation was checked with a symbolic-algebra implementation using SymPy and a separately written Python standard-library implementation. The reported results agree. These checks were produced within one verification; their agreement is not an additional independent human review.

The shared starting point includes the admissible domain, the nonnegative six-parameter representation (3), its component-term counting rule, and the invariant-line tests (4)–(6). From

these supplied preparations, the verification separately determined the minima, constructed witnesses, and proved the lower bounds. The specification also included published examples, so the verification was not blind to those known examples. Neither derivation includes a full-text comparison of the classification domains in Newton's and Lyagina's original papers.

8 Comparison and scope

The comparison in Table 1 uses coefficients rather than class numbers. In Table 1 of [1], the printed line-type labels in rows 14–16 differ from the types obtained from those rows' coefficients using (4)–(6). Section 4 of that paper gives

$$\dot{x} = (x - y)^2, \qquad \dot{y} = (y - 1)^2.$$

Its six terms give configuration $\{\mathrm{P}_1, \mathrm{H}, \mathrm{P}_2\mathrm{H}\}$, as can also be seen by exchanging the species in row 8 of Table 3. The text calls this class 12. We retain the explicit system and its computed configuration as the comparison, without proposing a corrected historical numbering convention. Four parabolic sectors and two hyperbolic sectors alone do not distinguish the line-type multisets being minimized.

Lemma 6(ii) of [1] contains a further boundary discrepancy: for $\eta_2 > 1$ it prints $\eta_1 < 1/\eta_2$, whereas its proof, equation (3.43), gives $\eta_1 \leq 1/\eta_2$. The admissible system $P = \frac{1}{2}u^2$, $Q = (u-v)^2$ has finite slopes $1/2, 2$ and attains equality. Its resultant is $1/4$ and its ratios are $-2, 4, 0$, including the vertical direction. The nonnegative parameter cone in (3) includes this valid boundary; no strict version of that source inequality is used here.

Related minimality problems use different constraints. Chen and Tang [4] classify minimal-dimensional degenerate zero-one networks; their two-dimensional condition concerns the stoichiometric subspace and their degeneracy condition differs from a zero planar Jacobian at a selected equilibrium. Plesa and Sprott [5] study small monomial counts for three-dimensional chemical systems with chaos. Beldiman and Halanay [6] classify minimal canard networks; their canard-point conditions require nonzero cross derivatives. These results do not supply the nine minima of Table 1 in the material examined.

The result covers the real coefficient domain of Definition 1. It leaves systems with zero resultant, equilibria on the boundary of the concentration quadrant, other molecularity constraints, and minimum numbers of reactions outside its scope. The bounded literature review has not located the same nine minima. Full-text review of the historical sources and a broader priority search remain incomplete.

Use of AI. AI tools assisted the algebraic search, certificate software, literature comparison, and drafting. The mathematical claims are supported by the proof and exact data described above. Responsibility for their review and any decision to publish rests with the author.

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